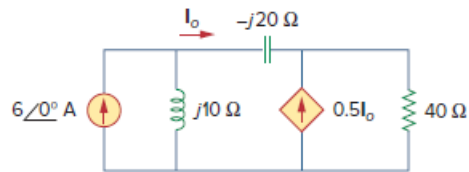


Problem

In the circuit of Fig. 11.40, determine the average power absorbed by the $40\text{-}\Omega$ resistor.

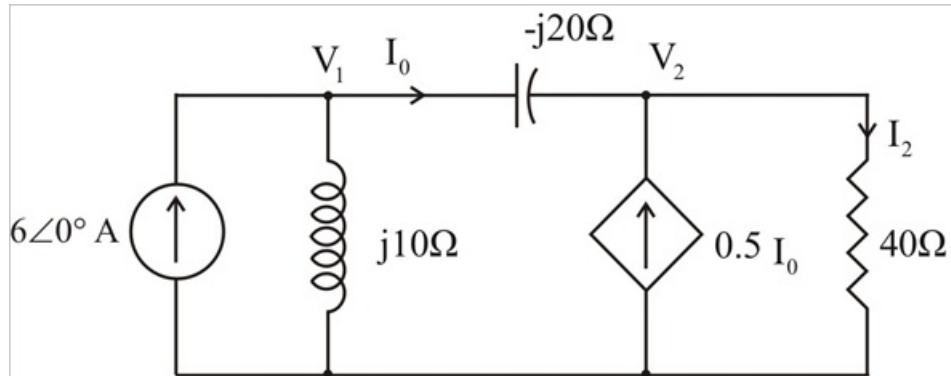
Figure 11.40:



Step-by-step solution

Step 1 of 4

We apply Nodal analysis to the following circuit.



Step 2 of 4

At node 1,

$$6 = \frac{V_1}{j10} + \frac{V_1 - V_2}{-j20}$$

$$6 = \frac{V_1}{j10} - \frac{(V_1 - V_2)}{j20}$$

$$V_1 = j120 - V_2 \dots\dots\dots (1)$$

At node 2,

$$0.5 I_0 + I_0 = \frac{V_2}{40}$$

$$1.5 I_0 = \frac{V_2}{40}$$

$$\text{But, } I_0 = \frac{V_1 - V_2}{-j20}$$

Hence,

$$\frac{1.5(V_1 - V_2)}{-j20} = \frac{V_2}{40}$$

$$3V_1 = (3 - j)V_2 \dots\dots\dots (2)$$

Step 3 of 4

Substituting (1) into (2),

$$j360 - 3V_2 - 3V_2 + jV_2 = 0$$

$$j360 - 6V_2 + jV_2 = 0$$

$$V_2 = \frac{j360}{6 - j}$$

$$V_2 = 59.18 \angle 99.46^\circ \text{ V}$$

Step 4 of 4

$$I_2 = \frac{V_2}{40}$$

$$I_2 = \frac{59.18 \angle 99.46^\circ}{40}$$

$$I_2 = 1.4795 \angle 99.46^\circ \text{ A}$$

Therefore average power absorbed by the 40Ω resistor is,

$$P_{40\Omega} = \frac{1}{2} |I_2|^2 R$$

$$P_{40\Omega} = \frac{1}{2} \left(\frac{9}{\sqrt{37}} \right)^2 (40)$$

$$\therefore \boxed{P_{40\Omega} = 43.78 \text{ W}}$$